

An attempt at plugging physics into this problem

“How do I read this?” So.. yes the file is structured quite poorly, but I’ve tried to meet the following constraints: the discussed topics are layed out in such a way that the most difficult-to-implement things appear closer to the beginning; also I’ve tried to retain some sort of readability (trying to introduce everything gradually not to induce some sort of undesirable effect). I don’t know how unreadable this is to a person who is seeing this stuff for the first time. I guess you’ll have to consult me on some topics.

WAKE EFFECT CONSIDERATIONS

When wind turbines are arranged in a straight line with a lateral offset, the first turbine directly facing the free-stream wind alters the flow field downstream. This aerodynamic interaction is governed by the wake effect, which impacts the power output, and efficiency of the subsequent turbine.

The first turbine extracts kinetic energy from the wind. This creates a low-velocity wake region directly behind its rotor. The wake spreads outward horizontally and vertically as it moves downstream. The expansion rate depends on ambient turbulence and atmospheric stability. High shear at the wake boundaries generates intensive small-scale turbulence. This increases the turbulent kinetic energy (TKE) reaching downstream rotors. Large-scale atmospheric eddies cause the wake to oscillate slowly from side to side. This creates dynamic, time-varying alignment variations with downstream turbines.

Because of the offset, subsequent turbines are not fully centered in the wake. They experience asymmetric flow, where one side of the rotor operates in the free-stream wind and the other side operates inside the slow, turbulent wake. Total power loss is significantly lower than a perfectly aligned inline configuration. The effective wind velocity facing downstream rotors is higher because they capture part of the unshifted free-stream air.

The first turbine can be deliberately rotated away from the wind direction (yawed). This introduces a lateral force that pushes the wake further away from the downstream turbines, maximizing total farm power.

The first turbine can operate at a reduced thrust coefficient (pitching blades or changing rotor speed). This allows more kinetic energy to pass through, speeding up wake recovery for the rest of the array.

HOW DO WE IMPLEMENT THIS?

To quantify the interaction between an upstream turbine (T_1) and a downstream turbine (T_2) in an offset array, we compute the shared area between the expanded wake profile of T_1 and the rotor plane of T_2 .

As the wake travels downstream over a distance x (the longitudinal spacing between turbines), its radius expands. Using the standard Jensen top-hat wake model, the wake radius $R_w(x)$ is calculated as:

$$R_w(x) = R_1 + k \cdot x$$

Where:

- R_1 is the rotor radius of the upstream turbine.
- k is the wake decay coefficient (should be somewhere around 0.075 in our case, **But we can play with this parameter**).
- x is the downstream distance along the wind direction.

Let d be the net perpendicular distance between the center of the expanded wake of T_1 and the hub center of T_2 at the downstream position. If the geometric lateral offset is Δy and the wind direction is perfectly parallel to the array line:

$$d = \Delta y$$

If the wind blows at an angle θ relative to the array line, d must be adjusted using trigonometry to reflect the true cross-wind tracking distance.

The overlapping area $A_{overlap}$ depends on the relationship between $R_w(x)$, d_{Genn} , and the rotor radius of the downstream turbine R_2 :

- **No Overlap** ($A_{overlap} = 0$): Occurs if $d \geq R_w(x) + R_2$. The downstream turbine is entirely in the free-stream flow.
- **Complete Submersion** ($A_{overlap} = \pi R_2^2$): Occurs if $R_w(x) \geq d + R_2$. The downstream turbine is fully inside the wake.
- **Partial Overlap** ($0 < A_{overlap} < \pi R_2^2$): Occurs if $|R_w(x) - R_2| < d < R_w(x) + R_2$. This matches your offset configuration.

For the partial overlap regime, the intersection area of two overlapping circles separated by center distance d is given by:

$$A_{\text{overlap}} = R_w(x)^2 \cos^{-1} \left(\frac{d^2 + R_w(x)^2 - R_2^2}{2d \cdot R_w(x)} \right) + R_2^2 \cos^{-1} \left(\frac{d^2 + R_2^2 - R_w(x)^2}{2d \cdot R_2} \right) - \frac{1}{2} \sqrt{(-d + R_w(x) + R_2)(d + R_w(x) - R_2)(d - R_w(x) + R_2)(d + R_w(x) + R_2)} \quad (1)$$

The dimensionless flow overlap fraction f_{overlap} , which directly scales the effective wind velocity deficit facing T_2 , is defined as:

$$f_{\text{overlap}} = \frac{A_{\text{overlap}}}{\pi R_2^2}$$

It should be noted that the wake effect not only reduces the average velocity of the wind (and so its kinetic energy), but also largely contributes to the increase in turbulence, hence the wind streams, which would be “more separated” in a laminar flow case, undergo quite a vigorous mixing as the flow is no longer laminar. So, the wake effect “averages” some parameters across the altitudes, like air humidity and air temperature. What do I mean by “average” – well... I don’t know, because turbulent flows are **impossible** to describe analytically – we’ll have to use our AI to somehow adjust for this.

To utilize this we’ll need to know the relative positions of turbines. I’ve found their coordinates and it seems like their orientation angle is irrelevant, since turbines utilized in this farm orient themselves to maximize performance. I’ll try to link the coordinates somewhere.

THE BETZ LIMIT. It establishes the absolute theoretical maximum efficiency that an ideal wind turbine can achieve. It dictates that no turbine, regardless of design, can capture more than $\frac{16}{27} \approx 59.3\%$ of the kinetic energy contained in an open wind stream.

An oversimplified intuition: Obviously, a wind turbine extracts energy by slowing down moving air.

- **100% Efficiency Case:** The turbine would have to stop the wind completely. If the wind stops entirely behind the rotor ($v_2 = 0$), no more air can pass through the system, and the airflow chokes.
- **0% Efficiency Case:** The turbine allows air to pass through without slowing it down ($v_2 = v_1$). No kinetic energy is extracted.

The Betz Limit solves the fluid dynamics optimization problem: finding the exact deceleration ratio that extracts maximum power while ensuring continuous mass flow through the rotor plane.

ANALYTICAL DERIVATION

Using 1D actuator disk theory based on the principles of Conservation of Mass and Conservation of Linear Momentum, the derivation proceeds as follows:

The mass flow rate ($\dot{m}$) of air passing through a rotor disk of area A per unit time is given by:

$$\dot{m} = \rho \cdot A \cdot v_{\text{avg}} \quad (2)$$

where ρ is the air density, and v_{avg} is the fluid velocity at the rotor plane. According to momentum theory, v_{avg} is the arithmetic mean of the undisturbed upstream velocity (v_1) and the downstream wake velocity (v_2):

$$v_{\text{avg}} = \frac{v_1 + v_2}{2} \quad (3)$$

The power (P) extracted from the wind corresponds to the rate of kinetic energy loss:

$$P = \frac{1}{2} \dot{m} (v_1^2 - v_2^2) \quad (4)$$

Substituting Equations (1) and (2) into Equation (3) yields:

$$P = \frac{1}{4} \rho A (v_1 + v_2) (v_1^2 - v_2^2) \quad (5)$$

Axial Induction Factor We introduce the axial induction factor (a), which parameterizes the wind velocity slowdown at the rotor plane:

$$v_{\text{avg}} = v_1 (1 - a) \quad (6)$$

From conservation of momentum, it follows that the downstream wake velocity is:

$$v_2 = v_1(1 - 2a) \quad (7)$$

Substituting Equations (5) and (6) into Equation (4) transforms the power equation into:

$$P = \frac{1}{2}\rho A v_1^3 \cdot [4a(1 - a)^2] \quad (8)$$

Maximizing the Power Coefficient The total available power in a wind stream of velocity v_1 passing through area A is $P_{\text{avail}} = \frac{1}{2}\rho A v_1^3$. The dimensionless power coefficient (C_p) represents the efficiency fraction:

$$C_p(a) = 4a(1 - a)^2 \quad (9)$$

To determine the maximum possible value of C_p , we compute the first derivative with respect to a and set it to zero:

$$\frac{dC_p}{da} = 4(1 - a)^2 - 8a(1 - a) = 0 \quad (10)$$

Factoring out $4(1 - a)$:

$$4(1 - a)[(1 - a) - 2a] = 0 \implies 1 - 3a = 0 \implies a = \frac{1}{3} \quad (11)$$

Theoretical Maximum Value Substituting the optimal induction factor $a = \frac{1}{3}$ back into Equation (8):

$$C_{p,\text{max}} = 4 \left(\frac{1}{3}\right) \left(1 - \frac{1}{3}\right)^2 = 4 \cdot \frac{1}{3} \cdot \frac{4}{9} = \frac{16}{27} \approx 0.5926 \quad (12)$$

Thus, an ideal wind turbine can extract a maximum of **59.3%** of the available kinetic energy.

Implementation idea

Modern commercial turbines operate at peak efficiencies between $C_p = 0.40$ and 0.50 due to wake turbulence, drag, mechanical losses, etc.

Normalization: Transform our target variable into the power coefficient domain:

$$\text{Target } C_p = \frac{\text{Actual Power Output}}{\frac{1}{2}\rho A v_1^3} \quad (13)$$

Force the-whatever-library-we-are-using to predict $C_p \in [0, 0.60]$. During model inference, reverse the operation by multiplying the model's output by the component $\frac{1}{2}\rho A v_1^3$.

Bounding: We need to set a boundary condition establishing the physical maximum upper bound for each specific timestep and environmental condition:

$$P_{\text{Betz}} = 0.593 \cdot \left(\frac{1}{2}\rho A v_1^3\right) \quad (14)$$

Then if any telemetry record indicates a measured power output greater than P_{Betz} , flag it as sensor failure, anemometer icing, or a database anomaly to remove it as invalid data point.

MODELING WIND FARM WAKE MIXING VIA TEMPERATURE GRADIENTS

The rate at which air layers mix after passing through a wind turbine (wake recovery) depends heavily on atmospheric stability. The temperature gradient determines this stability, which dictates the turbulent kinetic energy (TKE) available for mixing.

To use temperature and pressure data at different altitudes, calculate the potential temperature (θ) to eliminate compressible pressure effects:

$$\theta = T \left(\frac{p_0}{p}\right)^{R_d/c_p} \quad (15)$$

Where:

- T = Absolute temperature (K) at a given altitude.
- p = Atmospheric pressure (Pa) at that altitude.
- p_0 = Reference surface pressure.

Next there are some ideas which should be less difficult to implement.

SOME BASIC GAS PROPERTIES

At operational velocities the moving air masses can be considered to closely follow the ideal gas model. Wind power is directly proportional to localized air density ρ , which fluctuates based on temperature, elevation, and pressure.

$$P = \frac{1}{2} \rho A v^3$$

Extract ambient temperature (T) and barometric pressure (P_{atm}).

$$\rho = \frac{P_{\text{atm}}}{R_{\text{dry}} \cdot T}$$

$$v_{\text{corrected}} = v \cdot \left(\frac{\rho}{\rho_{\text{standard}}} \right)^{1/3}$$

This is aimed to liminate systematic overestimation during high-temperature, low-pressure periods and underestimation during cold, high-pressure regimes.

KINETIC ENERGY FLUX

Kinetic energy flux intersecting the rotor swept area scales non-linearly with the cube of wind speed (v^3). Here are somewhat plausible ideas:

1. Generate explicit polynomial features for velocity (v^2 and v^3).
2. Construct a composite theoretical available power metric:

$$P_{\text{available}} = v^3 \cdot \rho$$

WIND SHEAR AND POWER LAW PROFILE; BOUNDARY CONDITIONS

Turbines operate within the atmospheric boundary layer where velocity increases non-linearly with altitude.

Surface-level meteorological observations differ significantly from hub-height velocities.

The Hellmann power law provides a model for vertical wind shear profiles:

$$v = v_{\text{ref}} \left(\frac{z}{z_{\text{ref}}} \right)^{\alpha}$$

We need to estimate the atmospheric stability coefficient ($\alpha \approx 0.14$ for neutral stability, or somehow dynamically calculate it via our AI model.) Extrapolate measured anemometer velocity (v_{ref} at height z_{ref}) up to the specific turbine hub-height (z).

Turbine Inertia and Mechanical Lag.

Turbine drivetrains possess high rotational inertia. Blades obviously cannot instantaneously adapt to micro-scale velocity fluctuations.

Possible implementation Steps:

1. Compute rolling temporal lags for wind speed ($t-1, t-2, t-5$ intervals).
2. Apply something like an Exponential Moving Average to wind velocity to simulate the physical delay in aerodynamic acceleration.
3. Extract a discrete acceleration feature:

$$\Delta v = v_t - v_{t-1}$$

Lastly, comes the obvious:

Cut-in, cut-out conditions

Since this section is quite obvious and there is not much else to cover here, I will list the wind turbine characteristics:

SIEMENS GAMESA SG 3.4-132 or equivalently, **G132-IIA 3.465MW**

$$R_{\text{rotor}} = 66m; \quad L_{\text{blade}} = 64.5m; \quad S_{\text{swept-area}} = 13685m^2; \quad H_{\text{hub}} = 84m$$

$$V_{\text{cut-in}} = 3 \frac{m}{s}; \quad V_{\text{cut-out}} = 25 \frac{m}{s}; \quad V_{\text{rated}} = 10.3 \frac{m}{s}$$

The turbine coordinates have been found on OpenStreetMap.